

Influence of Local Site Effects in the Ahmedabad Mega City on the Damage due to Past Earthquakes in Northwestern India

by B. Sairam, A. P. Singh, Vandana Patel, Vasu Pancholi,
Sumer Chopra, V. K. Dwivedi, and M. Ravi Kumar*

Abstract The city of Ahmedabad of Gujarat in western India suffered severe damage during the 2001 M_w 7.7 Bhuj earthquake, despite being ~250 km away from the epicenter. Similar damage patterns were also reported during the 1819 Allah Bund earthquake (M_w 7.8). To investigate the probable causes, we employed an integrated approach using multichannel analysis of surface waves (MASW), single and array microtremor measurements, broadband earthquake data, and geotechnical investigations. Significant differences in site characteristics are observed in both the damaged and undamaged areas. The investigations revealed shear-wave velocity values in excess of 320 m/s and less than 220 m/s in the undamaged and damaged areas, respectively, in the top 6 m of the subsurface. The refusal ($N_{1(60)}$) values are observed at 6–20 m and 20 m depth onward in the undamaged and damaged areas, respectively. The amplification factor in the damaged areas varies from 3.3 to 6.6 in the 1.5–2.0 Hz frequency range. On the other hand, the amplification factor in the undamaged areas varies from 1.0 to 3.0 in the 0.7–1.5 Hz frequency range. Nevertheless, the damages were mostly restricted to mid-to-high-rise buildings, located in the western side of the Sabarmati River, where the presence of a paleochannel is reported, and in some southeastern parts of the city, along lakes and ponds. The low-rise buildings on the eastern side of the river in the old city area remained almost intact. Our study confirms that local site effects together with the poor quality of construction contributed to the damages.

Electronic Supplement: Table of shear-wave velocities, and figures of estimated shear-wave velocity and comparison of standard penetration test (SPT) $N_{1(60)}$ model at the damaged and undamaged areas.

Introduction

The influence of local soil conditions on the damage patterns is documented during the 1985 M_w 8.1 Michoacán earthquake that caused severe damage in the Mexico City, located about 350 km away from the epicenter (Furumura and Kennett, 1998). A similar destruction scenario was reported in the city of Ahmedabad, western India, during the 26 January 2001 M_w 7.7 Bhuj earthquake, located ~250 km away from the epicenter. This city is the most populated one in Gujarat and is categorized as a mega city in the country, because it has a population of more than six million. It is also inscribed as India's first World Heritage City by the United Nations Educational, Scientific and Cultural Organization. Incidentally, the damages observed due to the Bhuj earthquake were significantly different in various parts of the city

of Ahmedabad (e.g. Bhandari and Sharma, 2001; Goel, 2001; Earthquake Engineering Research Institute [EERI], 2002; Stewart *et al.*, 2002; Narayan and Sharma, 2004; Saito *et al.*, 2004). Most of the damaged buildings were mainly confined to the western and southeastern parts of the city and the undamaged areas to eastern part of the city (Fig. 1). The EERI team reported that the damage was primarily restricted to mid-to-high-rise buildings (5–12 stories) along the paleochannel of the Sabarmati River, lakes, and ponds (Fig. 1), while the low-rise buildings were left intact (EERI, 2002). Nonuniform damage patterns were also observed in the pre- and postsatellite images (Saito *et al.*, 2004). Interestingly, no damages were reported in the eastern part of the city, although the density of buildings was higher compared to that in the damaged area (Goel, 2001). It is worthwhile to mention that the eastern part of the city is older than the western part that was founded in 1411 (Gupta *et al.*, 2002).

*Now at Institute of Seismological Research, Raisan, Gandhinagar 382 007, Gujarat, India.

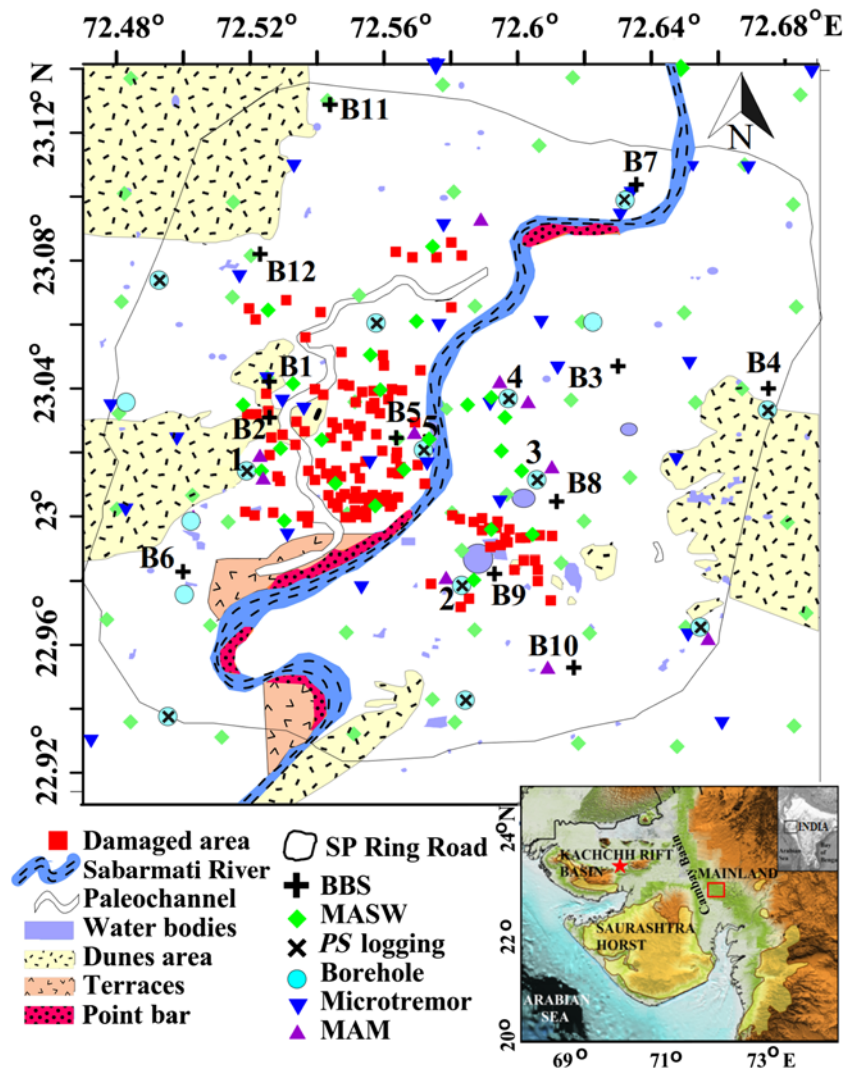


Figure 1. Geomorphological map of the study area with locations of geophysical and geotechnical survey sites. The damaged area is marked based on the location of buildings damaged during the 2001 Bhuj earthquake (modified after Bhandari and Sharma, 2001; Govindaraju *et al.*, 2004). The damaged buildings identified from satellite image (Saito *et al.*, 2004) and Earthquake Engineering Research Institute (EERI) report (EERI, 2002) are mainly confined to the western and southeastern parts of the city. The curvilinear line shows the boundary of the city. Site numbers (1–5) are used for detailed discussions. The sites B1–B12 show the location of broadband seismograph stations. (Inset) Location in the Gujarat region. The star shows the epicenter of 26 January 2001 M_w 7.7 Bhuj earthquake. The square shows the location of the city of Ahmedabad in the Gujarat map. MASW, multichannel analysis of surface waves; MAM, microtremor array measurement. The color version of this figure is available only in the electronic edition.

The old town located on the eastern side of the river mostly contains 1–2-story load-bearing masonry brick wall buildings, which virtually did not experience any damage, except thin cracks in the walls (EERI, 2002). However, the western and southeastern parts of the newly developed area, mostly comprising decade old reinforced concrete frame structures of 5 stories (ground + four) and a large number of 10-story buildings suffered severe damages (Goel, 2001; EERI, 2002; Murty *et al.*, 2002). About 130 buildings

of 5–10 stories were collapsed, resulting in ~750 casualties. Such extensive damages were not expected in the city of Ahmedabad because it is located far from the epicenter. The level of accelerations at such a large distance (Chandra *et al.*, 2002) may not be sufficient to cause severe damage to even poorly constructed buildings. Similar, unexpected high damages were also observed in the western parts of the city of Ahmedabad during the 1819 M_w 7.8 Allah Bund earthquake (Macmurdo, 1824; Bilham, 1999). Locations of the collapsed buildings (Fig. 1) clearly reveal that the damage pattern is not random. The damage occurred in some groups toward the western and the southeastern parts of the city. All the clusters consist of 5–8 buildings, ~1 km in diameter, with the distance between adjacent clusters varying between 1.5 and 2.5 km. Although the epicenter is located about W13°N from Ahmedabad (Fig. 1), the collapsed buildings are mostly aligned along W40°N, as shown in Figure 1. Furthermore, the EERI team did not observe any geological evidence for a subsurface connection that could give rise to such a damaged pattern. We, therefore, investigated the site response of the basin on which the town of Ahmedabad is situated. Several researchers advocated that local site effects contributed to the damages in Ahmedabad, besides poor quality of building constructions (e.g., Bhandari and Sharma, 2001; Goel, 2001; Murty *et al.*, 2002; Pande and Kayal, 2003; Govindaraju *et al.*, 2004; Parvez and Madhukar, 2006; Parvez and Rosset, 2014). However, to understand the causes of the damage patterns in the city of Ahmedabad no detailed studies have been carried out so far, except qualitative postearthquake investigations.

In the recent past, effects due to several earthquakes have established the fact that local site conditions have a significant role in the amplification of ground motion, especially in those areas located on unconsolidated young sedimentary materials (see Hunter *et al.*, 2002; Rastogi *et al.*, 2011; Sairam *et al.*, 2011; Singh, 2015). During strong earthquakes, the local site conditions can strongly influence the ground shaking and the associated damage to nonengineered structures (Campillo *et al.*, 1989; Aki, 1993; Hough *et al.*, 2002; Rastogi *et al.*, 2011; Verma *et al.*, 2014; Singh, 2015). Detailed geophysical, seismological, and geotechnical investigations can provide reliable estimates of the local

site effects (Andrus *et al.*, 2006; Chopra *et al.*, 2013; Singh *et al.*, 2014, 2017). The average shear velocity (V_S) of the top 30 m of soil (V_{S30}) is an indicator of the strength of the ground and is a critical parameter to evaluate the near-surface stiffness and characterize a given site (Borcherdt, 1970; Boore *et al.*, 1997; Castro *et al.*, 1997; Park and Elrick, 1998; Garofalo *et al.*, 2016; Foti *et al.*, 2017). The National Earthquake Hazards Reduction Program (NEHRP) assigned site classes (Building Seismic Safety Council [BSSC], 2004) based on V_{S30} values only. Several studies have found a correlation between amplifications and V_{S30} (e.g., Borcherdt, 1994; Sairam *et al.*, 2011). The parameters describing the site effects in seismic codes are expressed through soil characterization and spectral amplification factor.

In this maiden study, we tried to elucidate the causes of significant damage patterns in the city of Ahmedabad through an integrated approach. In the damaged and undamaged areas, we employed geophysical approaches such as multichannel analysis of surface waves (MASW), array and single-station records of microtremors, and broadband earthquake records. The estimated results are compared with geotechnical data to quantify the site effects.

Geology of the Study Area

Ahmedabad is a part of mainland Gujarat that lies within the Cambay basin, where the thickness of Quaternary sediments is about 350 m. The city is underlain by unconsolidated recent alluvial sediments that are deposited down to depths of 35–100 m, followed by ~250–300-m-thick Pleistocene sediments (Merh and Chamyal, 1993; Parvez and Madhukar, 2006; Hasan *et al.*, 2013; Parvez and Rosset, 2014). Our study area is distinctly divided into the eastern and western parts by the Sabarmati River (Fig. 1). On the western side of the river, a paleochannel was found ~5.0 km away, which gradually shifted to the present path (Goel, 2001). There were several lakes and ponds in the western and southeastern parts of the city. On the western side of the river, the unconsolidated materials derived from the sediments transported by the river are deposited in the top ~6 m. Geomorphologically, the town is located on the banks of the Sabarmati River. Therefore, floodplains cover the entire city of Ahmedabad, where silty sediments to sands are exposed on the surface. The alluvium mainly comprises paleodeltaic, fluvial, and Aeolian sediments, consisting of alternate layers of fine- to coarse-grained sand, gravel, and yellowish-brown sticky clay. The upper layers are mostly silty fine sands of Aeolian nature (Kumar *et al.*, 1998; Parvez and Madhukar, 2006). Thick sequences of tertiary formations lie beneath the alluvium layers (Kaila *et al.*, 1990).

Geophysical Surveys

PS Logging

We carried out PS logging measurements at five sites within the damaged and six in the undamaged areas (see

Fig. 1 for location). In this method, initially, the probe is lowered into a fluid-filled borehole to a desired depth, to acquire *P*- and *S*-wave data. Later, a pressure wave is generated in the fluid using the seismic source in the probe. The pressure waves are converted to *P* and *S* waves at the borehole walls, which travel through the geological formations. Later, these waves are converted back to the pressure wave at the borehole wall. Consequently, the geophones in the probe detect *P* and *S* waves, which are recorded by a data acquisition system (OYO Corporation, 2002). The V_S data were acquired at every 1 m interval from the surface to the bottom of the borehole. At each site, V_S down to a depth of 40 m was estimated.

Multichannel Analysis of Surface Waves

We performed MASW at 65 sites within the damaged and undamaged areas in the city of Ahmedabad. This technique utilizes the dispersive properties of Rayleigh waves for the estimation of a V_S profile of the subsurface. The entire process of MASW consists of three major steps, namely acquisition of multichannel seismic data, generation of the dispersion curve, and estimation of the V_S profile. We followed the recommendation of Park *et al.* (1999), to combine the MASW with the common mid-point (CMP) roll-along data acquisition procedure to obtain a laterally continuous 2D V_S profile. A 48-channel engineering seismograph was used to acquire data in this current investigation. The data were acquired using the standard CMP roll-along technique to achieve a continuous shot gather. Vertically stacked 20 impacts of a 30-kg weight drop on a metal plate were used as a source to generate seismic waves. These waves were recorded by 24 vertical-component 4.5 Hz geophones/receivers planted at every 2 m interval along the profile, which responded to frequencies from 1 to 90 Hz in a 24-channel seismic shot gather. A typical record of the shot gather is shown in Figure 2a. Acquisition parameters of the geometry are selected to optimize the imaging of subsurface layers down to more than 30 m depth. The signal-to-noise ratio of the processed data was more than 85 at each site. In the dispersion curve, the analyzed frequency was in the 4–50 Hz range. A typical dispersion curve is shown in Figure 2b. Each dispersion curve was individually inverted to obtain a V_S profile by setting the initial model and performing a least-squares minimization (Xia *et al.*, 1999), inbuilt in SurfSeis (Kansas Geological Survey [KGS], 2010). The initial velocity model is taken from the nearby PS logging measurements. *In situ* density measurements in nearby boreholes are also used in the initial model. All 1D V_S profiles are gathered in a sequential order to generate a 2D V_S section. An example of a 2D V_S model for site number 5 (see Fig. 1 for location) is shown in Figure 3.

At a few sites, where the MASW and PS logging points are collocated, the V_S values are estimated using MASW and PS logging for validating our results. The V_S values also correlate well with the geotechnical lithologs. We found that all the methods provide comparable results (Fig. 4). Finally, a

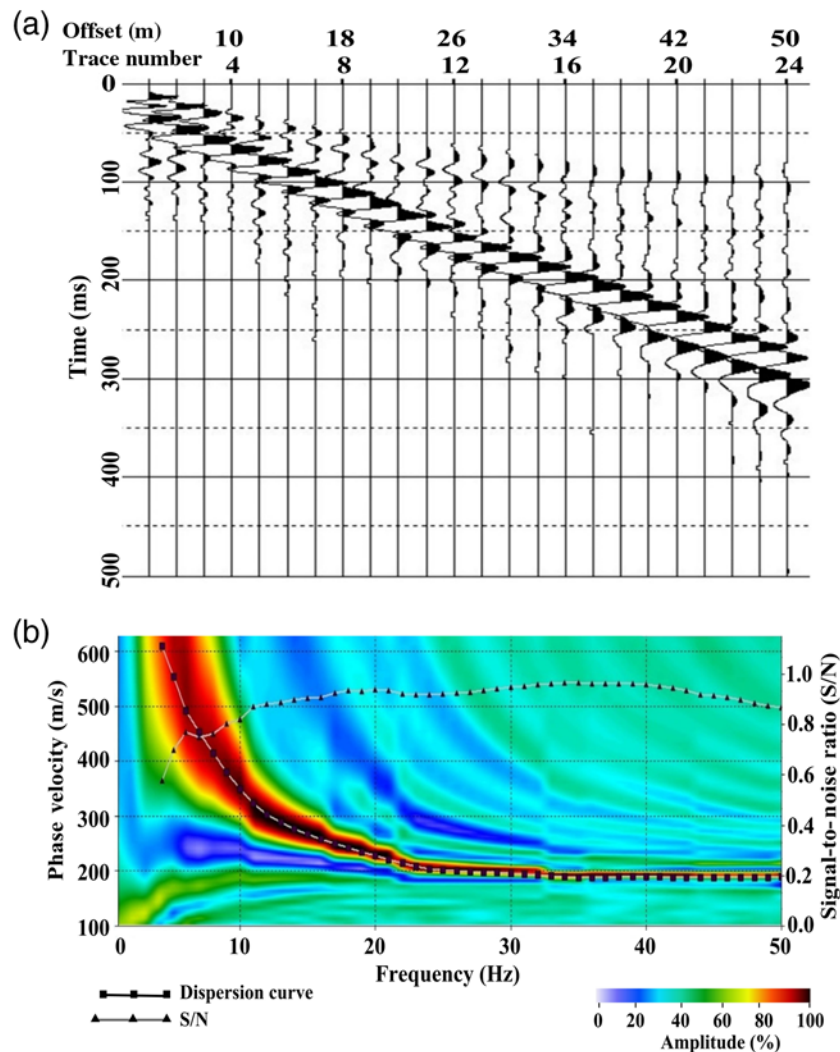


Figure 2. (a) A typical shot gather record of 24 channels, at site 5 (see Fig. 1 for location). The geophone interval is 2 m and the total spread length is about 50 m. (b) Rayleigh-wave dispersion curve in the 4.5–50 Hz frequency range obtained by processing the shot gather. The scale is shown at the bottom of the figure. The color version of this figure is available only in the electronic edition.

time-averaged velocity of the top 30 m (V_{s30}) of each site is estimated (Ⓔ Figs. S1 and S2 and Table S1, available in the electronic supplement to this article), and the site is classified following the NEHRP classes.

Geotechnical Investigations

In the city of Ahmedabad, a total of 16 boreholes (4 in damaged and 12 in undamaged areas) are drilled for determining the lithological variations and decipher local geological site conditions (Fig. 1). Different laboratory tests such as specific gravity, field dry density, natural water content, sieve analysis, hydrometer analysis, Atterberg limit, direct shear test, and free swell index are performed on soil samples, following the Indian Standards (IS-2131, 1981) for determination of the index properties of soils. Finally, lithologs are prepared for all the boreholes. Standard penetration tests

(SPTs) is conducted following IS-2131 (1981) in all the boreholes at every 3.0 m interval (Fig. 4). The field SPT N -values are corrected to obtain the standardized $N_{1(60)}$ values using the formula of Kayen *et al.* (1992) and the correction factor of Robertson and Wride (1997). The corrected N -values are referred as $N_{1(60)}$ because the original SPT hammer has about 60% efficiency (Skempton, 1986). The overburden pressure correction is done following the recommendations of Youd *et al.* (2001). Sample $N_{1(60)}$ models of the top 30 m of the damaged and the undamaged areas are plotted in Figure 5. The $N_{1(60)}$ models of all borehole are provided in Ⓔ Figure S3.

Microtremor Measurements

Single Station. We acquired single-station microtremor measurements at 60 locations in and around the city, in the early morning. The data were collected for 60 min at 100 samples/s. Three-component Lennartz LE-3D (5 s resonance period) seismometer coupled with a 24-bit City-shark II system was used in this survey. The sites 1–4 (see Fig. 1 for locations), which are closer to the boreholes, are considered for the analysis at undamaged and damaged sites. The horizontal-to-vertical (H/V) spectral ratio (HVSr) is obtained for all the single-station microtremor measurements (Nakamura, 1989). Although, a similar survey has been done by Parvez and Madhukar (2006) for seismic microzonation of the city of Ahmedabad, the measurement sites are slightly different than those in

the present study. We prepared resonant frequency maps of the region to understand the local site effects and estimate the thickness of soft sediments.

Microtremor Array Measurement. We have also conducted microtremor array measurement (MAM) at 10 sites in the city. The selection of MAM sites was made based on the availability of open space closer to damaged and undamaged areas (Fig. 1). The MAM at seven sites was performed using a 15 and 30 m triangular geometry. Additionally, a slightly larger aperture using 30 and 60 m was considered at three sites to explore deeper structures. In MAM, all the stations were preferably placed in a circular array (Garofalo *et al.*, 2016). We explored the V_s structure down to 100 m depth at site number 1 in the damaged area and at site numbers 3 and 4 in the undamaged areas for understanding the deeper seismic response attribute (Fig. 6). Seven sets of three

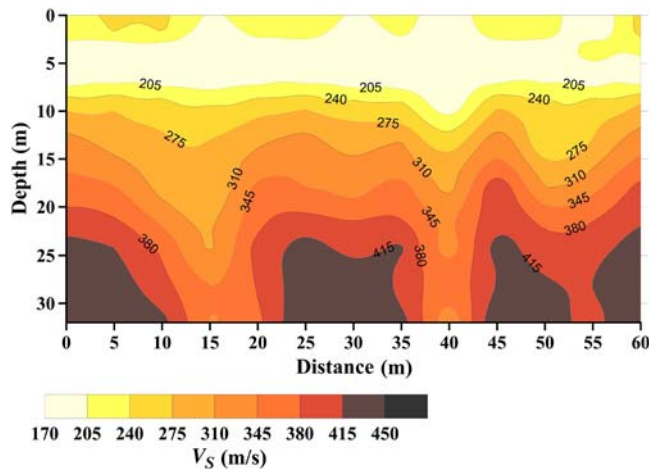


Figure 3. The 2D V_S model at site 5 estimated by the MASW method. At upper 8 m, V_S is ~ 200 m/s, while lower $V_S \sim 180$ m/s layer is observed at 5–7 m depth. The scale is shown at the bottom of the figure. The color version of this figure is available only in the electronic edition.

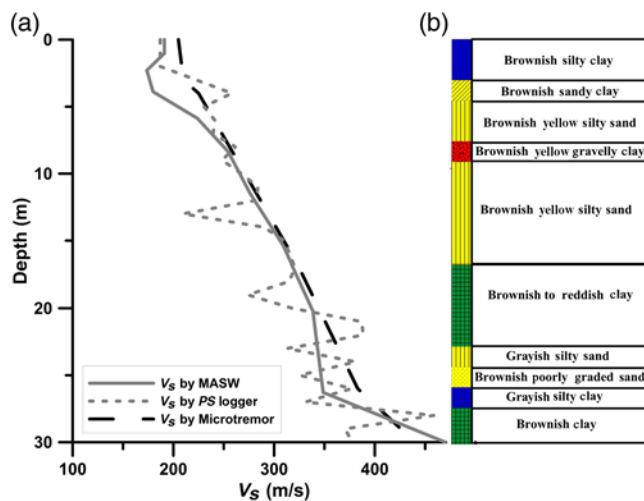


Figure 4. (a) A comparison of the shear-wave velocity (m/s) model, obtained from the PS logger, MASW, and MAM models; (b) lithologs down to depth of 30 m at site 5. The water table >20 m is observed in the city. Codes in the legend column are given following Indian Standard (IS-2002). The color version of this figure is available only in the electronic edition.

components Lennartz LE-3D seismometers each coupled with a Cityshark II system were used in this survey. The system has a master remote control that is used to trigger all the seven stations at the same time, to avoid any phase shift (Chatelain *et al.*, 2000). For reliable experimental conditions, the guidelines proposed by Koller *et al.* (2004) in the framework of Site EffectS Assessment using AMBient Excitations (SESAME; WP02 of SESAME Project), Garofalo *et al.* (2016), and Foti *et al.* (2017) have been followed.

For all the MAM sites, the fundamental-mode dispersion curves of Rayleigh waves were determined using

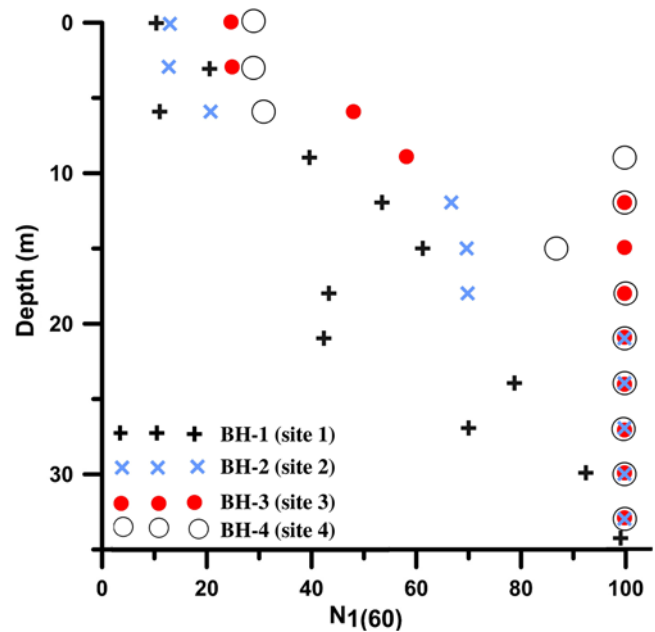


Figure 5. A close comparison of the SPT $N_{1(60)}$ model at the damaged and undamaged areas. The BH-1 and BH-2, and BH-3 and BH-4 are represented damaged and undamaged areas, respectively. The color version of this figure is available only in the electronic edition.

the frequency–wavenumber (f - k) analysis (Capon, 1969), as implemented in the Geopsy software package (Wathelet *et al.*, 2008). This method is based on the picking of energy maxima in the wavenumber plane for sliding time windows and narrow frequency bands. The method provides azimuth and slowness for the most coherent plane-wave arrivals. A wavenumber grid search is performed using an iteratively re-defined grid adjustment to allow for the accurate estimation of local maxima. The central frequencies of the frequency bands were sampled equidistantly on a logarithmic scale, and the frequency bandwidth was defined to be 6% of the corresponding center frequencies. The time-window length was chosen as 200 times the central period. The dispersion curves are extracted for each of the subarrays considering the resolution limits arising from the array geometry and combining the individual dispersion curves into one. In this study, we gathered the energy maxima for all the array sizes into one single histogram and picked the resulting dispersion curve, the minimum and maximum wavenumber limits for each subarray being determined from their theoretical array response. The dispersion curve was selected based on the approach of Wathelet (2008). The dispersion curves of smaller and larger subarrays were merged into a single curve, which was inverted using the Dinver software following the recommendations of Wathelet *et al.* (2008). The advantages and disadvantages of the f - k method are clearly described by the Garofalo *et al.* (2016) and Foti *et al.* (2017).

The phase slowness at larger aperture (60 m) MAM sites (sites 1, 3, and 4) shows a typical behavior. At all these sites, reliable dispersion curves are obtained between 1.0 and

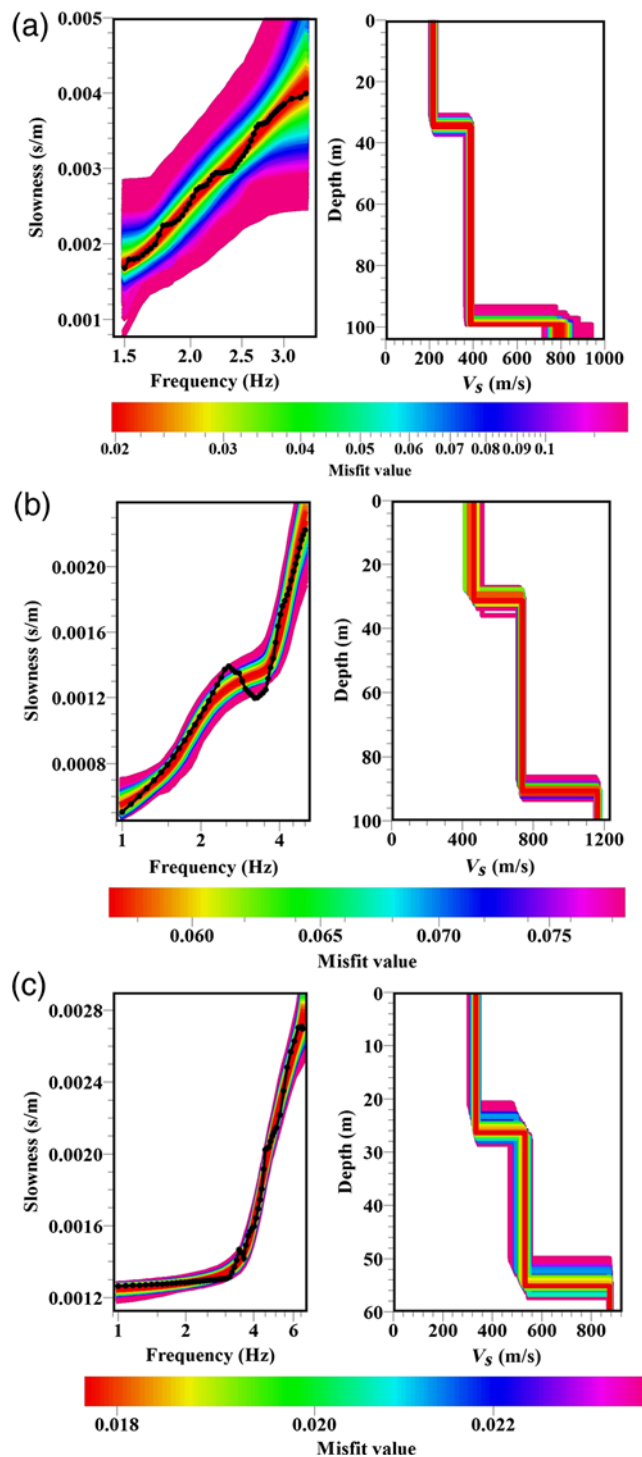


Figure 6. Results of MAM analysis at (a) damaged area (site 5) (b) undamaged area (site 3) and (c) undamaged area (site 4). The left-side part of plots show the dispersion curves at the sites (see Fig. 1 for locations). The right-side plots show the V_S profiles obtained from inversion of the dispersion curves. The observed dispersion curves are represented by black dots. The scales indicate the misfit of the models and the data. The MAM has revealed $V_S \geq 400$ m/s at 36 m depth in damaged; while $V_S \leq 600$ m/s in the undamaged sites. The color version of this figure is available only in the electronic edition.

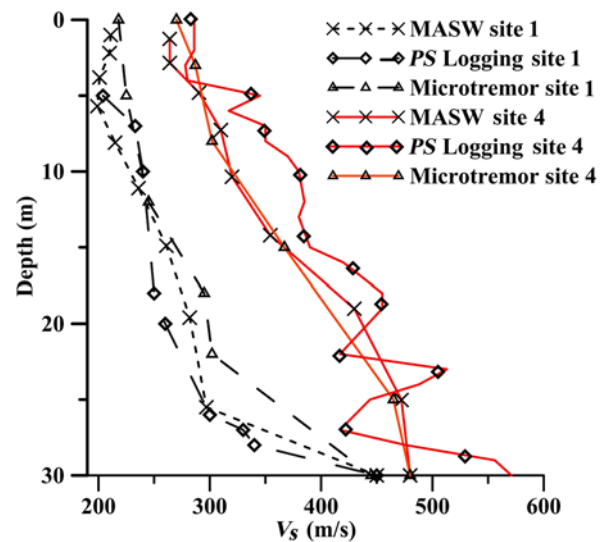


Figure 7. A close comparison of the V_S (m/s) model at the damaged and undamaged areas. V_S models in the damaged areas (sites 1 and 2) are shown as dashed lines; whereas V_S models in the undamaged areas (sites 3 and 4) are shown as thick lines. The color version of this figure is available only in the electronic edition.

6.0 Hz, and phase slowness between 0.0008 and 0.005 s/m. At the damaged sites, the dispersion curves are obtained at the 1.5–3.2 Hz frequency range with phase slowness between 0.0017 and 0.004 s/m. On the other hand, two sites that are outside the damaged area show the lowest slowness, between 0.0006 and 0.0028 s/m, in 1.0–6.0 Hz frequency range. In all the cases, the phase slowness increases as the frequency increases (Fig. 6). The damaged sites are characterized by very soft and thick sedimentary layers.

The inversion is performed using the neighborhood algorithm (Sambridge, 1999a,b). This algorithm is based on subdividing the solution space into Voronoi cells and refining the sampling in regions of low misfit during each iteration. The misfit is defined by the distance between the calculated and observed dispersion curve, at every frequency sampling point. When various modes are identified, inversion of all the modes is performed by employing a multimodel misfit (Wathelet *et al.*, 2008). A simple two-layer model based on MASW and *PS* logging observations at the respective sites is chosen as the starting model. The density in each layer was kept constant. The misfit values are indicated to judge the best-obtained velocity model out of the several models used for inversion. We compared the V_S estimated from MAM with those from MASW and *PS* logging, and found that they are well correlated at shallower depths (Figs. 4 and 7, and E Figs. S1 and S2). A spatial distribution map of V_{S30} was prepared, combining the MASW, MAM, and *PS* logging measurements (Fig. 8).

Analysis of Broadband Earthquake Records

In the city of Ahmedabad, the Institute of Seismological Research installed broadband velocity seismometers

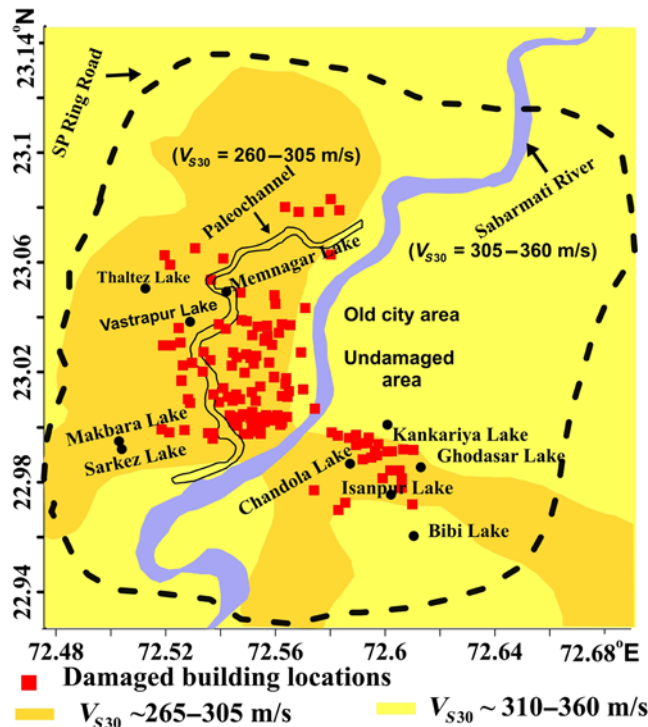


Figure 8. Map showing the V_{S30} variation in the study area. Squares are the locations of the damaged or collapsed buildings due to the Bhuj earthquake (Bhandari and Sharma, 2001; Govindaraju *et al.*, 2004). The color version of this figure is available only in the electronic edition.

(120 s–50 Hz frequency range) at 12 sites and recorded earthquake signals over a minimum period of one month at each site. The epicentral distances of the earthquakes are ~ 250 km. Recoding was continuous at the rate of 100 samples/s. We employed single-station estimates of HVSR using 24 local earthquake records ($M \geq 3.5$). A time window of 10.5 s beginning 0.5 s before the S -wave phase arrival was used to compute the fast Fourier transform (FFT). This time-window length was chosen such that it contains most of the S -wave energy. The FFT was applied to all the three components of earthquake records, and the amplitude spectra were smoothened following the technique of Konno and Ohmachi (1998). The estimated HVSR of 12 sites is shown in Figure 9. We compared the response of damaged (B1 and B2) and undamaged sites (B3 and B4) in detail (Fig. 10).

Comparison between V_S Profiles and Analysis of Uncertainties

We compared profiles estimated using different geophysical and geotechnical observations (Figs. 4 and 7, and ⑤ Figs. S1 and S2). These comparisons provided an opportunity to evaluate the accuracy of the employed approaches. The Jacobian matrix is used to estimate uncertainties on the dispersion curve, which is linked to the shear-wave velocity profile (Lai *et al.*, 2005; Foti *et al.*, 2017). It is well known

that due to the data uncertainties in active and passive methods and the nonlinearity of the problem itself, the solution of the observed dispersion curve inversion is generally nonunique (Garofalo *et al.*, 2016; Foti *et al.*, 2017). Thus, uncertainties in the determination of V_S are directly linked to the errors on the dispersion curves and to the frequency range over which the curve is retrieved. Uncertainty in the dispersion curve may also depend on the number of receivers. However, we have used recommended receiver configurations in MASW and MAM that are important for spatial sampling of the wavefield. On the other hand, in the MAM inversion, considering the misfit ranked set of inverted results, it is additionally possible to judge the uncertainty in the modeled parameters, usually represented in terms of the minimum misfit solution, often without any uncertainty estimates. Parameter uncertainties are underestimated when data error correlations are ignored and when V_P , V_S , and density parameters are fixed in the inversion. The corresponding depth curves in the MAM show layers at all depths being resolved with confidence. Additionally, the standard deviations for the HVSR peak frequencies deduced directly from the curves are ± 1 in all the cases.

It is found that uncertainty is lesser for the shallow layers, in terms of the average relative difference in V_S being $< 3\%$. However, the V_S values at deeper depths are significantly overestimated with large scattering (15%). These deviations mostly appear at deeper depths where very stiff formations with large impedance contrast do not allow the penetration of sufficient amount of the source energy. We noticed that the disparity among the different geophysical and borehole data is generally 10% for PS logging, 12% for MASW and MAM, and 13% for geotechnical approaches. A maximum of 20% disparity could be adopted as a realistic bound for the shear-wave velocity (Stephenson *et al.*, 2005). In addition, the uncertainties of the V_{S30} measurements using MASW are ± 30 m/s; whereas uncertainties of the V_S measurements in the MAM are observed similar to the MASW at shallower depths.

Results and Discussion

In this study, most of the HVSR peaks have more or less a bent shape, which indicates lack of high impedance contrast with the formations changing gradually as semiconsolidated material overlying the beds. A spatial distribution of the predominant frequency map was prepared, combining the estimated HVSR results of the microtremor and earthquake records (Fig. 11). The results show that the predominant frequency mostly varies in the 0.7–2.2 Hz range in the city. A few peaks are observed in the frequency bands of 4.0–10.0 Hz (Figs. 9 and 10). Variations in the predominant frequency confirm the differences in the nature of sediments. We believe that the difference in thickness may be due to erosion of soft sediments and deposition by rivers or lakes (Parvez and Rosset, 2014).

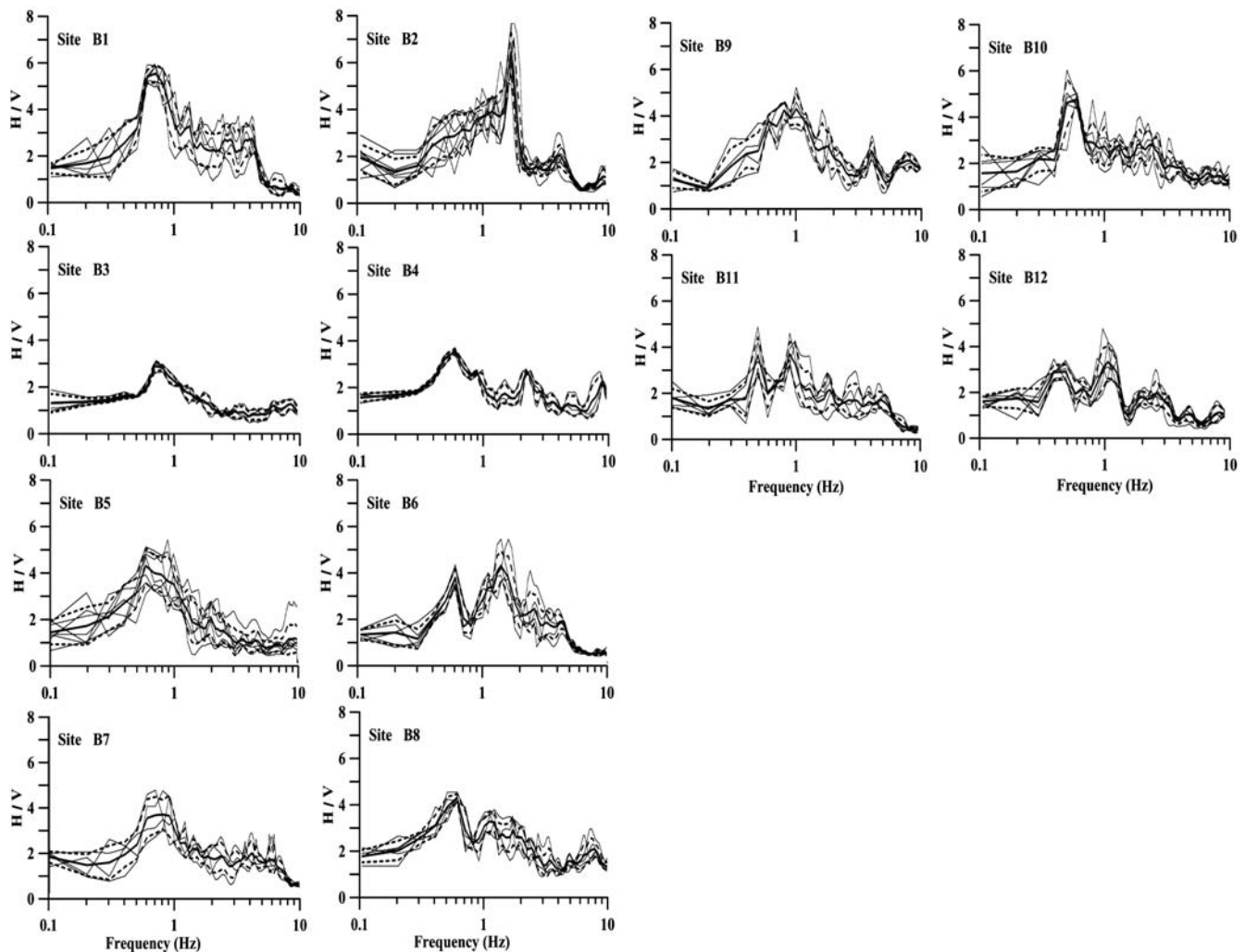


Figure 9. Horizontal-to-vertical (H/V) spectral ratio (HVSr) from earthquake records of the study area. Sites B1, B2, and B5 are in the damaged area, whereas sites B3 and B4 are in the undamaged area. Sites B6–B12 are at locations surrounding the damaged area. HVSrs show high amplification factors (3.0–6.6 times) in the 1.5–2.0 Hz frequency, which can resonate more than five-floor buildings.

Both earthquake and microtremor observations show high amplification factors (3.3–6.6 times) in the 1.5–2.0 Hz frequency, especially in the damaged areas (Table 1). However, the amplifications below a factor of 2 are observed in the 4–10 Hz frequency range. Amplification less than 2 may not be significant, in accordance with the [SESAME \(2004\)](#) guidelines. At such a large distance from the epicenter, the energy of high-frequency waves attenuates and the most energetic spectral component of seismic waves is contained in the lower frequencies, which are amplified due to thick sediments in the study region. Consequently, the high-rise buildings that have low-resonant frequencies are expected to suffer damages. Because most of the collapsed buildings had five and above stories, they might have been affected by the resonant frequencies. A few researchers have estimated the natural period of 5–11 story buildings in both the damaged and undamaged areas and found them to vary between 0.3 and 1.0 s (Table 2; e.g., [Kono and Tanaka, 2001](#); [Govindraj et al., 2004](#)), closer to the natural period of the sites. The

natural period of a building can be calculated using an elementary relationship $T = n/10$, in which T is the period and n is the number of stories ([BSSC, 1995](#)). This empirical formula is restricted to buildings not exceeding 12 stories, having a minimum story height of 3 m ([BSSC, 1995](#)). The estimated natural frequencies of 5–12 and 1–2 story buildings notionally correspond to the 0.8–2.0 Hz and 5–10 Hz ranges, respectively (Table 1), which is well corroborated with the observed natural periods of the buildings in the city of Ahmedabad. The observed and calculated data of the buildings reveal that the natural periods of buildings in the damaged area are different from those in the undamaged area.

The HVSrs show high amplification factors (3.0–6.6 times) in the 0.7–2.2 Hz frequency, which can resonate more than five-floor buildings. Consequently, the building vibration may be increased many fold due to resonance phenomena. We may also interpret that the mid-to-high-rise buildings in the city of Ahmedabad can resonate strongly

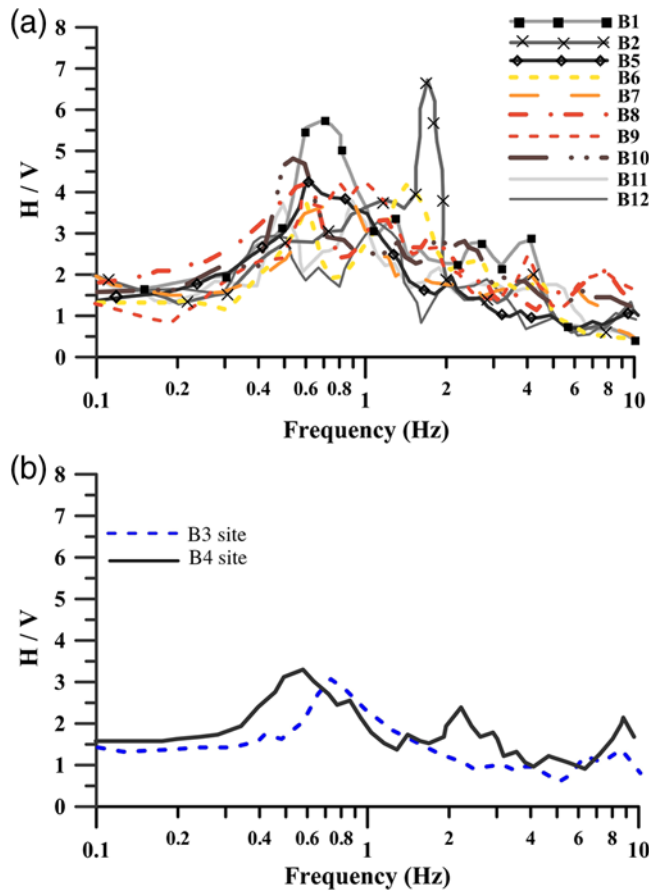


Figure 10. Comparisons of HVSR in the (a) damaged area (sites B1 and B2) and (b) undamaged area (sites B3 and B4). The color version of this figure is available only in the electronic edition.

with the ground vibrations and may experience more damages due to large earthquakes from the Kachchh region. One and two story buildings are less prone to resonance effects in the city due to far-field earthquakes.

The time-averaged velocity of the top 30 m (V_{S30}) was computed for each site (see details in ⑤ Figs. S1 and S2, and Table S1). Finally, a contour map of V_{S30} for the city of Ahmedabad is presented (Fig. 8). In this study, the point data are interpolated or extrapolated over the area of interest by the Kriging method. The V_{S30} value in the study area generally varies in the 260–360 m/s range. The area falls in the D-type soil according to the NEHRP classification. Additionally, geotechnical investigations at 16 sites down to 80 m depth show that the sandy and clayey layers are present alternatively from top to the termination depth (Fig. 4). We noticed that the top 6 m contains very loose fine-grained silty sand, followed by a layer from 6 to 26 m depth containing relatively compact silty sands with significant patches of silts that overlie a 7–15-m-thick strong clay layer. The upper 6 m layer contains very recent deposits, the second layer (6–26 m depth) has recent deposits, and the third (clayey

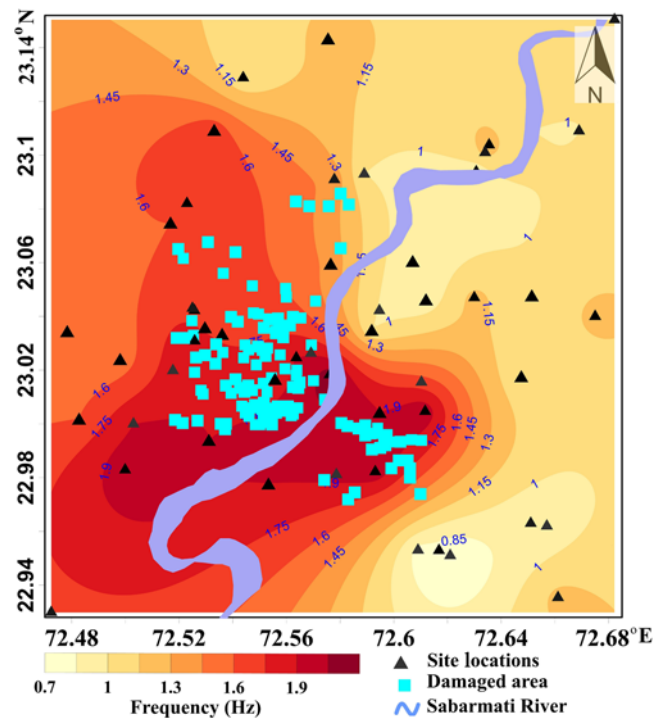


Figure 11. Spatial distribution of fundamental resonance frequency (Hz), the Sabarmati River is shown by curvilinear lines. Triangles show the measurement points; while the squares show the damage patterns. The frequency scale is shown in the bottom of each map. The color version of this figure is available only in the electronic edition.

layer) may represent a formation of Pleistocene (Merh and Chamyal, 1993). The depth to the water table is greater than 20 m in the city. Because the water table is slightly deeper, liquefaction may not be expected. Also, no case of liquefaction was reported during the past earthquakes in the city of Ahmedabad (Govindaraju *et al.*, 2004).

In the case of MAM analysis for larger aperture, we observed a lower $V_S \sim 220$ m/s down to 36 m depth in the damaged area, whereas a higher $V_S \sim 400$ m/s is seen in the outside area. A V_S value ≥ 400 m/s is encountered at 36 m depth in the damaged area, whereas it varies between 16 and 20 m depth outside. Furthermore, the V_S value is ≥ 800 m/s at about 90 m depth in the damaged area, whereas this value is reached between 40 and 56 m depth in the undamaged areas. The results suggest that the damaged sites may consist of softer soil deposits and/or filled sediments (e.g., Bhandari and Sharma, 2001; Goel, 2001; Parvez and Rosset, 2014). The results from MAM and MASW methods at shallower depths are in agreement, both in terms of layer depth and velocities (Figs. 4 and 7). The results are not comparable at deeper depths, because MASW has less probing potential compared with the MAM (Singh, 2015). The reliability of the inverted models has also been checked by means of forward modeling. A compact soil may be present at deeper depths beneath the damaged sites compared to the other sites.

Table 1
Comparison of Site Effects in the Damaged and Undamaged Areas of Ahmedabad

Parameters	Damaged Area	Undamaged Area
Area locations	The distribution of collapsed buildings is confined to certain pockets such as those along the western side of the Sabarmati river and along lakes and ponds	The old town is located on the eastern side of the river
Building types	Most of the buildings were of 5 stories and a few of 10 stories Natural frequencies of mid-to-high-rise buildings: mid-rise buildings, 5–12 stories: 0.8–2.0 Hz; high-rise buildings, 5 and above stories: below 2.0 Hz	Old city area has low-rise buildings Natural frequencies of low-rise, 1–2-story buildings may be in the 5.0–10.0 Hz range
Soil strength; $V_{S30}/N_{1(60)}$ values	V_{S30} ranges from ~265 to 300 m/s Lower $N_{1(60)}$ values generally found down to 30 m depth Refusal $N_{1(60)}$ encountered at 20–30 m depth and downward Top 20–30 m soil is weaker	V_{S30} ranges from ~310 to 360 m/s Higher $N_{1(60)}$ values generally found down to 30 m depth Refusal $N_{1(60)}$ encountered at 6–12 m depth and downward Soil is stronger
Site-response estimates	Amplifications in different frequency ranges: 5.6–6.6 in 0.6–2.0 Hz; 2.0–3.0 in 2.1–4.9 Hz; 1.0–2.0 in 5.0–10.0 Hz	Amplifications in different frequency ranges: 3.0–3.3 in 0.6–2.0 Hz; 1.1–2.4 in 2.1–4.9 Hz; 1.0–1.4 in 5.0–10.0 Hz

Table 2
Natural Frequencies of Mid-to-High-Rise Damaged Buildings

Serial Number	Name of Building	Number of Stories	Height (m)	Natural Frequency (Hz)	Damage Level due to Earthquake
1	Akshar Deep	5	14.2	1.49	4
2	Siddhi Flat	5	15.6	1.79	3
3	Mansi Complex	11	30.7	1.02	3
4	—	4–11	—	1.5–3	3

Source: Kono and Tanaka (2001) and Govindraju *et al.* (2004).

The V_S profiles down to 100 m, at a damaged site A1 and undamaged sites A2 and A3, resulting from the inversion of MAM data are shown in Figure 6, with a scale indicating the misfit value (see Fig. 1 for site locations). These indicate a remarkable difference in the soil structure. All the models are fairly consistent with minimum error or misfit. The shallow V_S may be statistically correlated with deeper V_S (Boore *et al.*, 2011). Although a tertiary layer down to depth of about 1000 m is observed, the depth of the actual bedrock is not very well known in the city. However, a study of the seismicity and tectonics of the Gulf of Cambay indicated that the graben comprises up to 5 km of post-Mesozoic clastic and shallow marine deposits that overlie the Deccan Trap basalt flows (Mukhopadhyay and Arora, 1999). They reported the thickness of basin sediments near Ahmedabad to be around 3 km.

The V_{S30} values in the damaged areas are in the 260–305 m/s range (Fig. 8), compared to 305–360 m/s in the undamaged areas (Table 3). The old city area of Ahmedabad has the highest V_{S30} values in the 340–360 m/s range. Low V_S values are observed in places covered by ancient lakes, ponds, and paleochannels. Thus, the damage scenario may be due to the presence of much younger and unconsolidated Holocene sediments that were the key factors, in addition to the resonance effect. In the damaged areas, the

major contrast in V_S was observed at a depth of 26 m. In the undamaged areas, V_S increases gradually with depth and no significant contrast is observed between the layers. In the damaged areas, the average V_S of the top 26 m layer is about 260 m/s. Our estimates of the predominant frequency (f) due to the contrast at 26 m depth (h), estimated using the formula $f = V_S/4h$, reveal a predominant frequency of ~2.5 Hz. More layers with contrast in shear velocity may exist at deeper depths also, because our HVSr estimates from broadband seismograph and microtremor data also show predominant frequencies at 2.5 Hz and below (Fig. 11). One of the reasons for the damages of buildings could be amplification and resonant frequency due to layers at 26 m and deeper depths. In the damaged areas, the soil at the top 30 m depth is D-type. However, in the undamaged area, the top 15 m is of D-type, followed by a C-type soil. The comparison of site attributes in the damaged and undamaged areas of Ahmedabad is summarized in Tables 1 and 3.

The lithologies of the study area show significantly different characteristics of the soil in the damaged and undamaged areas. A comparison of the corrected SPT N -values, $N_{1(60)}$ of the damaged and undamaged areas, reveals that the $N_{1(60)}$ values of the soil strata in the damaged areas are less than those in the undamaged areas (Fig. 5). In the damaged area, the top 6 m layer shows $N_{1(60)}$ values that are in the 10–20

Table 3
Comparison of Subsurface Layers and Their Properties in the Damaged and Undamaged Areas

Depth (m)	Geology	Age of the Formation	Damaged Area		Undamaged Area	
			Average V_S (m/s)	N -Value	Average V_S (m/s)	N -Value
6	Silty sand	Very recent	200–220	10–20	320–400	30–40
36	Consolidated clay sand	Recent sediments (Merh and Chamyal, 1993; Hasan <i>et al.</i> , 2013)	220–300	30–70	400–450	50–100
47	High plastic clay	Middle Pleistocene (Merh and Chamyal, 1993)	300–360	70–100	450–550	100
100	Highly consolidated clay	Lower Pleistocene	600–800	100	600–800	100
350	Tertiary	Tertiary (Hasan <i>et al.</i> , 2013)	900–1200	100	900–1200	100

Here, the depth refers to the bottom of the layer. Results from multichannel analysis of surface waves, PS logging, and microtremor array measurement are considered.

range, whereas in the undamaged area, the $N_{1(60)}$ values are 30–40. In the damaged area, the refusal $N_{1(60)}$ values ($N_{1(60)} \geq 100$) are encountered from 20 m onward (Fig. 5). In contrast, in the undamaged areas, the refusal $N_{1(60)}$ values are encountered at 6 m depth onward. In the damaged area, the density of soil in the top 20 m depth is in the 1400–1700 kg/m³ range, whereas in the undamaged area, the Bulk density is more than 2000 kg/m³. Lower $N_{1(60)}$ and density values indicate that the soil of the damaged area is weaker in strength than the soils of the undamaged area. The amplifications of ground motions are high in the areas of low V_S and $N_{1(60)}$, which may affect the built environ (Bendick *et al.*, 2001; Bhandari and Sharma, 2001; Goel, 2001; Hough *et al.*, 2002).

Some researchers reported that the damaged buildings had several structural defects such as open ground story, short column, poor quality of the construction, soft story, and irregularities in the buildings that lead to large damage (Goel, 2001). However, such buildings are expected to be randomly located and therefore cannot give rise to the observed clustered damage. Our integrated study shows that the site amplification at the predominant frequency coinciding with the natural frequencies of the damaged buildings, in addition to the loose unconsolidated soil in the damaged areas, contributed to the heavy damage, besides the poor construction practices.

Conclusions

We present results of integrated geophysical and seismological investigations together with geotechnical studies in the city of Ahmedabad and attempt to understand the causative factors for the damage patterns due to past earthquakes. HVSr curves reveal that the amplification factor is approximately two times higher in the damaged area, with the predominant frequency (0.7–2.2 Hz) matching with the mid-to-high-rise buildings. Therefore, the mid-to-high-rise buildings could have resonated with much-amplified ground vibrations and experienced heavy damage during the 2001 Bhuj earthquake. The estimated shear-wave velocities show

$V_S \sim 320$ and 220 m/s at 6 m in the undamaged and damaged areas, respectively. Our estimates show a good correlation with the site characteristics and damage pattern, suggesting that the strength of soil layers at varying depths is a dictating factor for damaged scenarios.

Data and Resources

We used the data acquired by the Institute of Seismological Research (ISR) Gandhinagar, Gujarat, India (www.isr.gujarat.gov.in, last accessed February 2017). Shear-wave velocities were estimated using multichannel analysis of surface waves (MASW), PS logging, and microtremor methods. Geometrics Geode Seismograph was used to acquire the data for MASW. PS logger of OYO Inc., Japan, was used for PS logging. A set of seven Lennartz LE-3D seismographs with Cityshark II system was used in microtremor surveys. Surfseis software of Kansas Geological survey, United States, was used for the processing of the MASW data. Broadband seismographs of Güralp, United States, were used to acquire the earthquake data. Many of the figures are produced using the Generic Mapping Tools software (Wessel and Smith, 1998).

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Institute of Seismological Research (ISR)
 Raikan
 Gandhinagar 382007
 Gujarat, India
 apsingh07@gmail.com
 (B.S., A.P.S., V.Pat., V.Pan., S.C., V.K.D.)

CSIR-National Geophysical Research Institute
 Hyderabad 500 007
 Telangana, India
 (M.R.K.)

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